Different Simulators & Emulators

TABLE I: Network simulators/emulators

Category	Tools	Granularity	Technology	Networking	Open Source
Discrete-event simulator	StarPerf [26]	Flow-level	STK [25] & Python & Matlab	Route calculation	✓
	Hypatia [27]	Packet-level	ns3 [30]	Routing protocol simulation	✓
	Yan *et al.* in [28]		OMNeT++ [31]		×
	Cheng *et al.* in [29]		ns3 [30] & Matlab		×
Virtual-network emulator	Mininet [32]	Real stacks	Virtualization (Namespace)	NA	Not for SN
	Etalon [37]		Container (Docker)		
	LeoEM [34]		Mininet [32]	Route calculation	✓
	StarryNet [35]		Container (Docker)	Routing software in containers/VMs	Partially
	Pan *et al.* in [36]		Container (Docker)		×
	Celestial [38]		MicroVM (firecracker [39])		✓
	NEaaS [40]		Virtual Machine (KVM [41])		×
	LORSAT [42]		Virtual Machine (VMWare)	NA	×
	OpenSN		Container (*Better Manager*)	Routing software in containers	✓

TABLE II: Performance comparison of SN emulators

SN Emulator	Emulation Efficiency	System Scalability	Function Extensibility
LeoEM	Efficient for SN emulation thanks to the lightweight virtualization of Mininet	Good vertical scalability on single machine, not horizontally scalable to more machines	Not able to emulate SN running distributed software (e.g., routing) due to the lightweight virtualization
StarryNet	Inefficient due to direct interaction with Docker CLI in SN emulation	Good horizontal scalability to multiple machines, but is not vertically scalable due to direct interaction with Docker CLI	Allow for distributed software (e.g, routing), but not flexible to modify images, topology, etc
OpenSN	Efficient due to our improvement on the process of using Docker CLI in our managers	Good vertical and horizontal scalability due to improvement on the process of using Docker CLI in SN emulation	Allow for distributed software (e.g, routing), and flexible to extend (due to separation of user configuration from container network management)

Check out: https://arxiv.org/html/2604.04498v1#S3

Q1 Journal, Pg 3: https://www.sciencedirect.com/science/article/pii/S1389128626003956?ref=pdf_download&fr=RR-2&rr=a2ade1e11ef041c5